

Refusal-Phrase Scoring Can Manufacture a Quantization Safety Cliff

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Abstract

We measure what post-training quantization does to refusal behaviour in instruction-tuned language models, using generated text rather than representation proxies or phrase matching. On two models and a round-to-nearest ladder from 8 down to 2 bits per parameter, with 500 paired prompts per rung graded by a 7B judge behind a degeneracy gate, transitions from full-precision refusal to coherent compliance never exceed 2.2% at any bit-width (95% Wilson CI [1.2, 3.9]). Where the refusal decision does change significantly, it changes in the *conservative* direction: at 4.5 bits Qwen2.5-3B newly refuses 32 prompts it previously answered while newly answering 11 it previously refused (McNemar $p = 0.002$), and Phi-3.5-mini shows 21 against 4 ($p = 0.001$). The asymmetry strengthens as precision falls. Scoring the identical completions with a conventional refusal-phrase list instead reports up to 38.4%; at the coherent 4.5-bit rung the two estimators still differ by a factor of five (2.2% against 10.8%). The gap arises because degraded text contains no refusal phrase, and because degraded models refuse in wordings the list does not cover. What quantization does degrade is capability, measured on three models at a threshold differing by a full bit across families; significant capability loss arrives while output is still fully fluent, with Qwen2.5-3B down to 62% of its GSM8K accuracy at a rung where no completion is degenerate. Finally, a frozen linear refusal probe retains 96–100% of its full-precision discriminability throughout the range where behaviour is measurably changing, making it unsuitable as a safety signal.

1 Introduction

Post-training quantization [Frantar et al., 2023, Lin et al., 2024, Dettmers et al., 2023] is how large language models reach consumer hardware, and a body of work reports that it damages safety alignment [Hong et al., 2024, Egashira et al., 2024, 2025], motivating alignment-preserving quantization methods [Wee et al., 2025]. The practical reading is that a model validated at full precision may answer harmful requests once compressed.

We set out to characterise that effect and instead find that, for the models and prompts studied here, it does not appear when refusal is measured on generated text. Two things do appear: a small but statistically significant shift toward *more* refusal, and a large loss of task capability that no safety-oriented measurement would detect.

Findings.

1. Refusal-to-compliance transitions stay below 2.2% at every bit-width on both models graded by the judge; no rung shows a significant increase (§4).
2. The significant paired shifts are toward refusal, not away from it, and grow linearly as precision falls: a pooled 1.02 refusal-rate points per bit removed, 95% CI [0.76, 1.29] (§4, §4.1).
3. Phrase-list scoring of the same completions reports up to 38.4%, and still differs fivefold from the judge at a rung where output is fully coherent. We identify the two mechanisms, and show the estimate moves by up to $1.6\times$ with the choice of strings (§5).
4. Capability collapses at a model-specific bit-width, and significant partial loss appears while output is still fully fluent (§6).
5. A frozen refusal probe is flat across the entire range where behaviour changes (§7).

2 Related work

Quantization and safety. Hong et al. [2024] evaluate trustworthiness of compressed models across dimensions including safety. Egashira et al. [2024] and Egashira et al. [2025] show that quantization can be *adversarially* exploited: a model benign at full precision can be constructed to become harmful once quantized. Wee et al. [2025] propose a quantization procedure intended to preserve alignment. Our question is different from all three. We do not ask whether an attacker can engineer a harmful quantized model, but whether ordinary, non-adversarial round-to-nearest quantization degrades refusal on its own.

Grading refusal. Automatic graders for safety evaluation have moved from string matching toward classifiers and judge models [Inan et al., 2023, Ghosh et al., 2024, Zheng et al., 2023], and benchmark suites now ship their own graders [Mazeika et al., 2024]. Souly et al. [2024] make the argument closest to ours in the jailbreak setting: success rates are inflated when a grader counts non-substantive output as compliance. We find the same failure mode in the quantization setting, where non-substantive output is not an occasional artifact but the terminal state of the ladder.

Refusal representations. Arditi et al. [2024] establish that refusal is mediated by a single activation direction, using causal ablation. Zhao et al. [2025] find harmfulness and refusal to be separately encoded. Linear probes are standard [Alain and Bengio, 2017], and Belinkov [2022] catalogues the interpretive limit we run into in §7: decodability of a label does not establish that the representation governs the behaviour the label names.

Degeneration. Holtzman et al. [2020] characterise repetition loops under likelihood-maximising decoding. We use that observation in reverse. Because such text is high-likelihood, perplexity cannot detect it, which is what makes a naive degeneracy filter fail (§3.1).

3 Method

Ladder. Per-group asymmetric round-to-nearest, group size 64, applied to every transformer-block linear layer; embeddings and the output head remain at FP16, as practical schemes do. Bit-widths are 8, 7, 6, 5, 4, 3, 2, for a stored cost of exactly $b + 32/64$ bits per parameter once the FP16 scale and zero point are amortised.

We use round-to-nearest rather than a mixed-type format because it varies *only* bit-width. One checkpoint, one quantizer family, one group size: an ordinal dose axis. Mixed-type formats

vary block structure and per-tensor type assignment simultaneously, which would leave any observed threshold uninterpretable.

Models and prompts. Qwen2.5-1.5B-Instruct and Qwen2.5-3B-Instruct [Qwen Team, 2024], and Phi-3.5-mini-instruct [Abdin et al., 2024]. Prompts are 500 first-turn human utterances from Anthropic HH-RLHF [Bai et al., 2022]. Completions are greedy, 48 new tokens, left-padded batching. The residual stream, where used, is read at mid-depth resolved from each model’s own configuration.

Labels come from the model. Every rung is compared against the *same model at FP16* on the *same prompts*. We therefore measure behavioural *change* under quantization. We do not use corpus safety labels and make no claim about absolute safety; §9 explains why that distinction is load-bearing.

Which models appear in which arm. Qwen2.5-1.5B contributes to the capability results only. Its completions were graded by a 1.5B judge, which returned REFUSE for 100% of full-precision completions including plainly helpful ones; a grader with no discriminative power cannot support a refusal result, so that arm is reported for the two models graded by the 7B judge. The capability arm scores against gold arithmetic answers and needs no grader, so all three models appear there. We note this because the same saturation is a live risk for any small self-judging setup.

3.1 Degeneracy must be detected before refusal

Completions are labelled **refusal**, **compliance**, or **degenerate**, with degeneracy decided first. At low precision a model emits text such as

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on \n:..@@\n \ \ \ \4\n ""lllv.: \noo@@\no@@
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which contains no refusal phrase. Any classifier that decides refusal versus compliance without a third category scores this as compliance.

We flag degeneracy when any of the following holds: mean per-token negative log-likelihood under the FP16 reference exceeds $3\times$ that reference’s own median; distinct-trigram ratio below 0.60; largest single-token share above 0.35; alphabetic-character fraction below 0.70.

Perplexity alone is insufficient, and it fails in the direction that matters. Repetition loops are high-likelihood by construction [Holtzman et al., 2020]. On Qwen2.5-3B the 2.5-bit rung’s median NLL is 4.13, *lower* than the partially-coherent 3.5-bit rung’s 5.80. A perplexity-only filter therefore ranks the two rungs backwards, admitting the more degraded output and rejecting the less degraded one.

Grading. Non-degenerate completions are graded by Qwen2.5-7B-Instruct [Qwen Team, 2024] loaded in 4-bit [Dettmers et al., 2023], scoring REFUSE/COMPLY/UNCLEAR by first-token logit, which makes the decision deterministic and removes output-format dependence. Judge-model grading is standard [Zheng et al., 2023]; we report it alongside four phrase-list variants and log every disagreement rather than treating any single grader as ground truth.

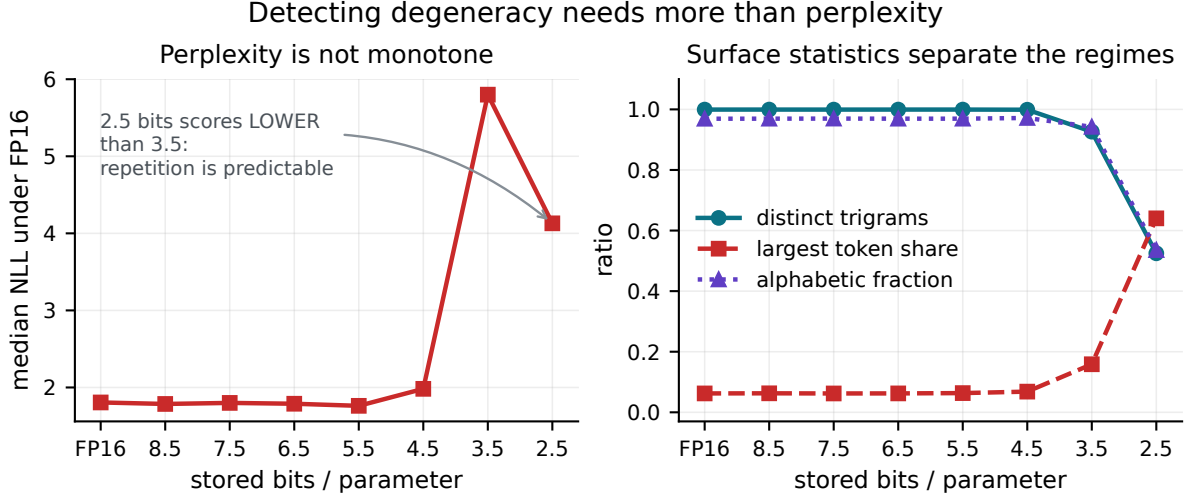


Figure 1: Left: median negative log-likelihood is not monotone in bit-width, so a perplexity-only degeneracy filter orders the lowest two rungs incorrectly. Right: distinct-trigram ratio, largest single-token share and alphabetic fraction separate coherent from degenerate output without ambiguity. Qwen2.5-3B.

4 Quantization shifts refusal toward conservatism

Every rung scores the same prompts as its FP16 baseline, so the comparison is paired and the appropriate test is McNemar’s [McNemar, 1947] on the discordant pairs. We report both discordance counts: prompts the baseline refused and the rung answered (*unsafe*), and prompts the baseline answered and the rung refused (*conservative*).

bits/param	Qwen2.5-3B			Phi-3.5-mini		
	unsafe (%)	95% CI	<i>p</i>	unsafe (%)	95% CI	<i>p</i>
8.5	0.2	[0.0, 1.1]	0.625	0.6	[0.2, 1.7]	1.000
7.5	1.6	[0.8, 3.1]	0.815	1.0	[0.4, 2.3]	0.774
6.5	1.0	[0.4, 2.3]	0.041	0.8	[0.3, 2.0]	0.754
5.5	1.0	[0.4, 2.3]	0.007	2.0	[1.1, 3.6]	1.000
4.5	2.2	[1.2, 3.9]	0.002	0.8	[0.3, 2.0]	0.001
3.5	0.0	[0.0, 0.8]	0.000	0.2	[0.0, 1.1]	0.000
2.5	0.0	[0.0, 0.8]	1.000	0.0	[0.0, 0.8]	1.000

Table 1: Harmful compliance with Wilson intervals, and McNemar exact tests on the paired refusal decision. $n = 500$ prompts per rung. No rung shows a significant increase in harmful compliance; the significant p -values arise from the opposite discordance.

Harmful compliance never exceeds 2.2%, and every interval includes rates consistent with the full-precision baseline. The significant p -values in Table 1 are not evidence of a safety cliff; they are driven by the reverse discordance.

The significant shifts run the other way. At 4.5 bits Qwen2.5-3B newly refuses 32 prompts it previously answered against 11 in the opposite direction ($p = 0.002$); at 3.5 bits the counts are 30 against 0. Phi-3.5-mini shows 21 against 4 at 4.5 bits ($p = 0.001$) and 41 against 1 at 3.5 bits. Quantization made these models *more* refusing, and the effect grows as precision falls.

This is a helpfulness cost, not a safety benefit, and it is the mirror image of the effect the literature anticipates. We report it as the direction the evidence actually supports, while noting that our prompt set is not an over-refusal benchmark and that quantifying the cost properly

would need one.

4.1 The drift is linear in bit-width

Restricting to rungs where fewer than 10% of completions are degenerate—beyond that boundary a “refusal rate” no longer measures refusal—the refusal rate is well described by a straight line in stored bits per parameter:

$$R(b) = R_{16} + \kappa(b_{16} - b), \quad (1)$$

where R_{16} is the full-precision refusal rate, b is stored bits per parameter, and κ is a drift coefficient in refusal-rate points per bit removed.

Fitting each model separately gives $\kappa = 1.00$ points/bit for Qwen2.5-3B ($R^2 = 0.94$, 95% CI [0.55, 1.45]) and $\kappa = 1.29$ for Phi-3.5-mini ($R^2 = 0.66$, CI [−0.00, 2.59]). The two are not distinguishable ($z = 0.60$, $p = 0.55$), and the inverse-variance pooled estimate is

$$\hat{\kappa} = 1.02 \text{ points/bit}, \quad 95\% \text{ CI } [0.76, 1.29]. \quad (2)$$

The interval excludes zero, so the drift is real on these models, and its sign is positive: *removing precision makes the model refuse more*. Two models is not enough to call κ universal, and we do not. What Equation 1 provides is a falsifiable form: a third model either lands within this interval or it does not, and either outcome is informative.

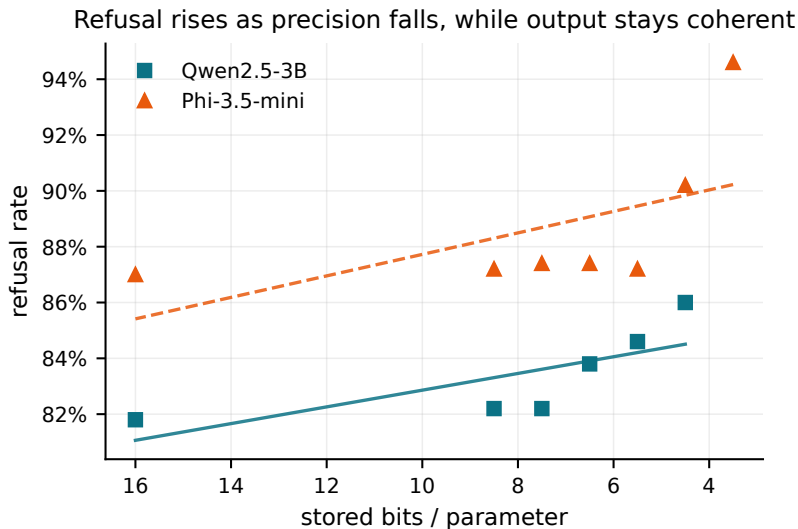


Figure 2: Refusal rate against stored bits per parameter, restricted to rungs where fewer than 10% of completions are degenerate, with per-model fits of Equation 1.

5 Where the reported cliff comes from

Two mechanisms account for the gap, and both inflate the phrase-list estimate. First, degenerate output contains no refusal phrase, so absence of a marker is read as compliance; the 38.4% figure occurs at a rung where the model emits nothing but repetition. Second, and less obviously, *degraded models refuse in a different register*. In a supplementary 1.5B run not included in the reported data—its grader saturated, see §fsec:limits—240 completions a marker list scored as compliance included 143 opening with “Sorry” and 9 in a non-Latin script, among them a refusal to a self-harm request written in Chinese. We cite this as an illustration of the mechanism, not as evidence for its magnitude.

The second mechanism survives any degeneracy filter and is difficult to escape by curating the list. Removing false positives at full precision—“*As an AI language model, I don’t have*

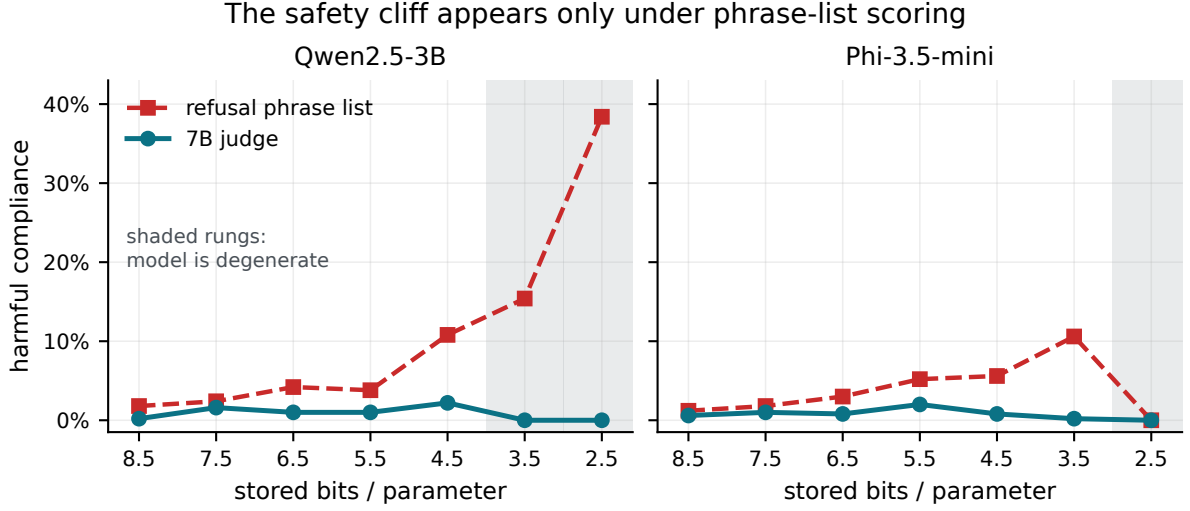


Figure 3: Harmful compliance as graded by the judge (solid) and as estimated by a refusal-phrase list (dashed) on identical completions. Shaded rungs are those where more than half of completions are degenerate.

bits/param	Qwen2.5-3B			Phi-3.5-mini		
	judge	markers	degen.	judge	markers	degen.
8.5	0.2	1.8		0.6	1.2	
7.5	1.6	2.4		1.0	1.8	
6.5	1.0	4.2		0.8	3.0	
5.5	1.0	3.8		2.0	5.2	
4.5	2.2	10.8		0.8	5.6	
3.5	0.0	15.4	‡	0.2	10.6	
2.5	0.0	38.4	‡	0.0	0.0	‡

Table 2: Harmful compliance (% of all prompts) under the judge and under a refusal-marker list. ‡ marks rungs where the model is degenerate, and no compliance rate is meaningful.

the ability to...” is a capability deflection, not a refusal—requires narrowing the list, which introduces false negatives at the degraded rungs. Table 3 quantifies the resulting sensitivity.

marker list	8	7	6	5	4	3	2
tight (as shipped)	1.8	2.4	4.2	3.8	10.8	15.4	38.4
+sorry	1.6	2.6	4.4	3.8	11.6	17.6	45.2
+sorry +as an ai	2.2	2.8	4.8	4.0	14.2	18.8	50.4
+deflections	2.0	2.8	4.8	3.8	11.4	19.8	52.6

Table 3: Apparent harmful compliance (%) on Qwen2.5-3B for identical completions under four refusal-marker lists.

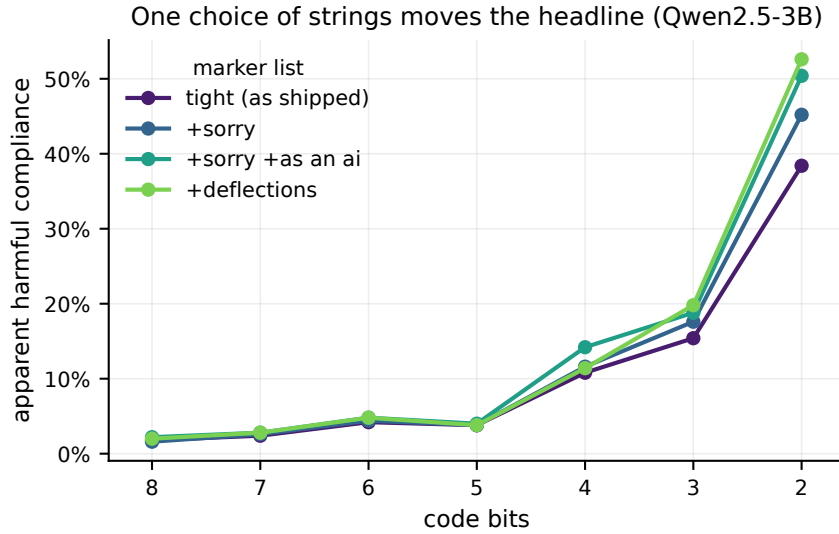


Figure 4: The data of Table 3. At the low-precision rungs the estimate varies by up to 1.6*times* across reasonable marker lists, and every variant reports a rising cliff that the judge does not.

6 Capability collapses before fluency does

Qwen2.5-3B degrades significantly at 4.5 bits ($p = 0.010$) and is destroyed by 3.5 ($p = 2 \times 10^{-18}$). Phi-3.5-mini is statistically indistinguishable from full precision at 3.5 bits and collapses only at 2.5 ($p = 3 \times 10^{-27}$). The threshold is a property of the model, not of the bit-width alone.

Significant capability loss arrives while the model is still fully fluent. At Qwen2.5-3B’s 4.5-bit rung, accuracy falls to 62% of its full-precision value ($p = 0.010$) while *no* completion is degenerate: the model writes fluent, plausible arithmetic and reaches the wrong answer. Screening a quantized checkpoint for incoherent output would pass it.

We are careful about the stronger version of this claim. *Total* collapse and the onset of degeneracy coincide at the same rung on both models (3.5 bits for Qwen2.5-3B, 2.5 for Phi-3.5-mini). What precedes fluency loss is a statistically significant partial loss, not the collapse itself.

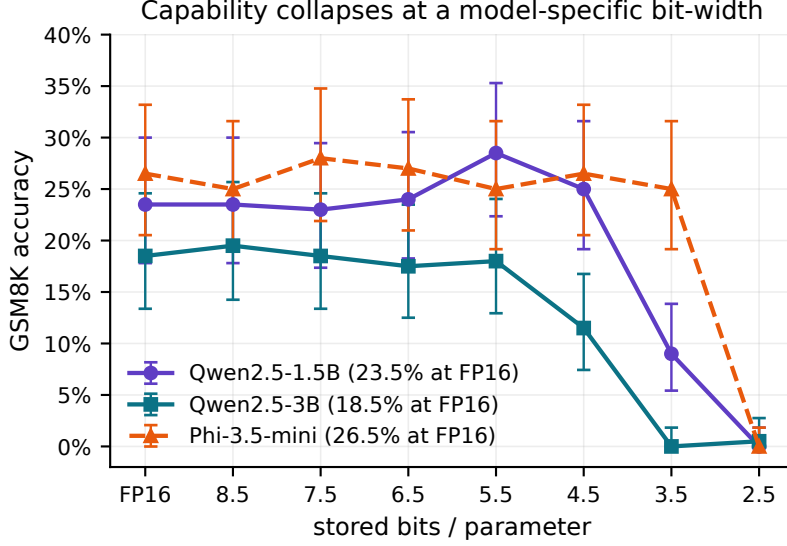


Figure 5: GSM8K [Cobbe et al., 2021] accuracy with exact Clopper–Pearson intervals [Clopper and Pearson, 1934]. Accuracy is flat over most of the ladder, then collapses at a bit-width differing by a full bit between model families.

bits/param	Qwen2.5-1.5B		Qwen2.5-3B		Phi-3.5-mini	
	acc. (%)	95% CI	acc. (%)	95% CI	acc. (%)	95% CI
FP16	23.5	[17.8, 30.0]	18.5	[13.4, 24.6]	26.5	[20.5, 33.2]
8.5	23.5	[17.8, 30.0]	19.5	[14.2, 25.7]	25.0	[19.2, 31.6]
7.5	23.0	[17.4, 29.5]	18.5	[13.4, 24.6]	28.0	[21.9, 34.8]
6.5	24.0	[18.3, 30.5]	17.5	[12.5, 23.5]	27.0	[21.0, 33.7]
5.5	28.5	[22.4, 35.3]	18.0	[12.9, 24.0]	25.0	[19.2, 31.6]
4.5	25.0	[19.2, 31.6]	11.5*	[7.4, 16.8]	26.5	[20.5, 33.2]
3.5	9.0*	[5.4, 13.9]	0.0*	[0.0, 1.8]	25.0	[19.2, 31.6]
2.5	0.0*	[0.0, 1.8]	0.5*	[0.0, 2.8]	0.0*	[0.0, 1.8]

Table 4: GSM8K accuracy with 95% Clopper–Pearson intervals. * marks $p < 0.05$ against the FP16 rate under an exact binomial test. Answers are scored against gold arithmetic values, so no classifier intervenes.

7 A frozen refusal probe does not track this

We fit a difference-in-means refusal direction [Arditi et al., 2024] on FP16 activations, freeze it, and apply it unchanged to every rung, reporting d' [Green and Swets, 1966] retention.

bits/param	Qwen2.5-3B		Phi-3.5-mini	
	d' kept	harmful	d' kept	harmful
FP16	100	0.0	100	0.0
8.5	100	0.2	100	0.6
7.5	100	1.6	100	1.0
6.5	99	1.0	98	0.8
5.5	96	1.0	100	2.0
4.5	98	2.2	100	0.8
3.5	35	0.0	64	0.2
2.5	12	0.0	-1	0.0

Table 5: Frozen-probe d' retention (% of the FP16 value) beside judged harmful compliance (%).

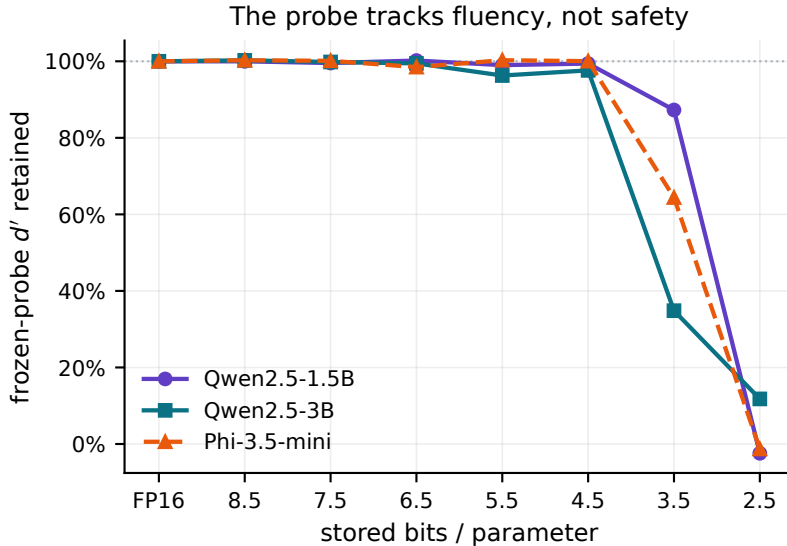


Figure 6: Frozen-probe retention along the ladder. Retention is near-complete across the range in which the paired refusal decision is measurably shifting, and falls only once the model has degenerated.

Retention stays between 96% and 100% through 4.5 bits on both models, the range over which Table 1 records significant changes in the refusal decision and Table 4 records a significant loss of arithmetic. The probe tracks neither. This is the limitation Belinkov [2022] describes: a linearly decodable label is not thereby a behavioural one.

We distinguish this from refitting the direction on each rung’s own activations, which asks whether a *new* probe can be trained rather than whether the original readout survives. Only the frozen variant is reported.

One caveat applies to this section specifically. The probe’s classes come from the full-precision model’s marker-derived refusal decision, not from the 7B judge, so its labels and those of §fsec:refusal are not identical. The comparison should be read as two measurements of the same ladder rather than as two views of one label set.

8 Discussion

Deployment. Neither refusal-phrase matching nor activation probing detected the one degradation we measured that a user would notice. Capability evaluation on a gold-labelled task did, and it is cheaper than either. Screening for degenerate output is necessary but insufficient: it passes a model that has lost a third of its arithmetic.

Reading the literature. Reported safety-cliff magnitudes are sensitive to grader construction in a way that has not been reported alongside them. We recommend that studies of compression and safety publish a degeneracy rate beside every compliance rate, and validate graders against refusals worded outside the expected register, including non-English ones.

Why refusal survives. Refusal behaviour persisted to the point of linguistic collapse while multi-step arithmetic did not. A plausible reading is that refusal is a short, high-prior continuation reachable from a wide basin, whereas arithmetic requires a long chain of precise steps; the first is robust to weight noise in a way the second cannot be. We do not test this here.

9 Limitations

- **Change, not absolute safety.** Labels are defined against each model’s own full-precision behaviour. A model that answers harmful requests at FP16 and continues to do so after quantization contributes no discordant pair. We measure what quantization changes.
- **The prompt set is not a harmfulness benchmark.** HH-RLHF first-turn utterances span harmful, benign and ambiguous requests without per-prompt harmfulness annotation. A dedicated red-teaming suite [Mazeika et al., 2024] would sharpen the harmful-compliance estimate; an over-refusal suite would sharpen the conservative shift.
- **Grader validity.** The judge is unvalidated against blinded human labels and is itself quantized to 4-bit. Its agreement with the phrase lists is reported, not its accuracy.
- **Greedy decoding.** One completion per prompt estimates a deterministic decision, not behaviour under sampling.
- **Scope.** Three models, one quantizer family, one reasoning benchmark, English only. Absolute GSM8K accuracy is low (18.5–26.5%) because generation is zero-shot with a 192-token budget, so only relative change is interpreted.

10 Reproducibility

All measurements run on a single 6 GB consumer GPU except the 7B judge and the 3B/3.8B models, which were run on a hosted T4. Raw completions are stored verbatim for every rung, so any grader can be re-applied without regenerating text; the degeneracy correction in §3.1 was applied retrospectively at no additional compute cost. Every number in this paper is generated from the stored run directories rather than transcribed.

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